CODE SPORT

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In this month's column, we will explore natural language processing.

We have been discussing many interview questions in the last few columns. This month, let's take a break from that and go back to our focus on natural language processing.

Last week, the premier conference in natural language processing — ACL 2020 (Association of Computational Linguistics) took place virtually for the first time *(https://acl2020.org/)*. There were a number of tutorials, workshops, keynotes and, of course, the research paper presentations. I would encourage our readers to visit ACL's anthology page where all the papers are available in full, to pick out and read the papers you find interesting. The videos should also be available on the ACL website soon.

In this month's column, I want to briefly talk about some of the interesting papers from the conference. ACL had 3429 submissions, out of which 779 papers were accepted. We will not be able to cover all of them here, so I just wanted to cherry pick a few and discuss those that I found interesting. I will cover a few this month, and some in next month's column.

Before we get to discussing the papers, ACL 2020 also had a number of keynotes, workshops and tutorials. Prof. Kathleen McKeown delivered a keynote titled, 'Rewriting the Past: Assessing the Field through the Lens of Language Generation'. She covered the past, present and future of NLP using language generation. While deep learning has impacted the adoption of NLP in many tools and tasks, it has not solved all the problems. It is not enough to merely show performance on some task, but also to understand the input and output in terms of: (a) whether this test data set is meaningful and representative enough for this task, and (b) is the model solving the actual task instead of solving the given data set. These are the two most important factors in determining the success of an NLP model in the real world. Unless the test data set we used to evaluate the performance of the NLP model is representative of the real world task, the model will not perform well in a real-world scenario.

Second and, most importantly, we need to figure out whether the model we have built is solving the actual task represented by the data set used to train it, or is it solving the data set only, instead of the actual task. This problem occurs because data sets often contain spurious correlations or artefacts that models can exploit, and hence models can perform well on a specific data set using these artefacts. However, real-world data will not contain such artefacts, and hence model performance suffers in the real world. We will discuss this in greater detail a bit later in this column.

Getting back to Prof. McKeown's keynote, she stressed that we should not be chasing leader board performance on specific data sets, but instead focus on applying NLP to specific and meaningful tasks.

ACL 2020 had a theme paper section that featured papers describing the progress and challenges in a specific area such as Natural Language Understanding (NLU), generalisation, and the linguistic diversity of current NLP models/tools. The entry that won the Best Theme Paper Award this year was 'Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data' by Prof. Emily Bender and Alexander Koeller. This paper talks about the limitations of the current state-of-art models in Natural Language Understanding (NLU).

Today, large pre-trained language models like BERT and GPT-2 are widely used in NLP/NLU. But do they really understand meaning? This award-winning paper makes the claim that any system trained in the language model task is trained on the surface form of the text, and not on the meaning represented by the text, and hence cannot successfully comprehend